

RMO– 2015(3)

Max Mark – 102

Time – 3 Hrs

Instructions:

- Calculators (in any form) and protractors are not allowed.
 - Ruler and compass allowed.
 - Answer all the questions. All Questions carry equal marks.
 - Answer to each questions starts on a new page. Clearly indicate the question number.
1. Two circles Γ and Σ with centres O and O' , respectively, are such that O' lies on Γ . Let A be a point on Σ and M mid-point of the segment AO' . If B is the point on Σ different from A such that AB is parallel to OM , show that the midpoint of AB lies on Γ .
 2. Let $P(x) = x^2 + ax + b$ be quadratic polynomial with real coefficients. Suppose there are real numbers. Suppose such that $\langle P(-1)^2, P(0)^2, P(1)^2 \rangle$ is an arithmetic progression of integers. Prove that a and b are integers.
 3. Show that there are infinitely many triples x, y, z of integers such that $x^3 + y^4 = z^{31}$.
 4. Suppose 36 objects are placed along a circle at equal distances. In how many ways can 3 objects be chosen from among them so that no two of the three chosen objects are adjacent nor diametrically opposite?
 5. Let ABC be a triangle with circum circle Γ and incentre I . Let the internal angle bisector of $\angle A$, $\angle B$ and $\angle C$ meet Γ in A', B', C' respectively. Let $B'C'$ intersect AA' in P and AC in Q , and let BB' intersect AC in R . Suppose the quadrilateral $PIRQ$ is a kite; that is $IP = IR$ and $QP = QR$. Prove that ABC is an equilateral triangle.
 6. Show that there are infinitely many positive real numbers a , which are not integers such that $a(a - 3\{a\})$ is an integer. (Here $\{a\}$ denotes the fractional part of a . For example $\{1.5\}=0.5$, $\{-3.4\}=0.6$.)

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